

02YMECH - Homework 7

Assigned for the week of Nov 3, 2025

Question

A particle of mass m moves under a central force $f(r)$. Its trajectory in polar coordinates is observed to be

$$r(\theta) = \frac{a}{1 + b \cos(\theta)},$$

where $a > 0$ and $0 < b < 1$.

1. Determine the central force $f(r)$ as a function of r .
2. State whether this force is attractive or repulsive for $0 < r < a$.